

Experiment-13

Aim

To perform Perspective Transformation on a video using OpenCV in Python by mapping the original frame into a new perspective using a transformation matrix.

Procedure:

1. Import the required libraries (cv2 and numpy).
2. Open the input video using cv2.VideoCapture().
3. Read each frame from the video.
4. Define four source points on the original frame.
5. Define four destination points for the transformed frame.
6. Compute the perspective transformation matrix using cv2.getPerspectiveTransform().
7. Apply the transformation using cv2.warpPerspective().
8. Display both the original and transformed video frames.
9. Press 'q' to stop the video.
10. Release the video object and close all windows.

Code:

```
import cv2

import numpy as np

cap = cv2.VideoCapture("video.mp4")

while cap.isOpened():

    ret, frame = cap.read()

    if not ret:

        break
```

```

rows, cols = frame.shape[:2]

pts1 = np.float32([[50, 50], [cols-50, 50],
                   [50, rows-50], [cols-50, rows-50]])

pts2 = np.float32([[0, 0], [cols-100, 50],
                   [100, rows-50], [cols, rows]])

M = cv2.getPerspectiveTransform(pts1, pts2)

transformed = cv2.warpPerspective(frame, M, (cols, rows))

cv2.imshow("Original Video", frame)

cv2.imshow("Perspective Transformed Video", transformed)

if cv2.waitKey(25) & 0xFF == ord('q'):
    break

cap.release()

cv2.destroyAllWindows()

```

Output

- The original video is displayed in one window.
- The second window displays the perspective transformed video, where the selected region appears viewed from a different angle.
- The transformation is applied continuously to every frame until the video ends or 'q' is pressed.

Result

The perspective transformation was successfully applied to the video using OpenCV. Each frame was transformed using a perspective transformation matrix, producing a new viewpoint of the original video.